

2025 Ch H2 Q2

Section: Chemical Changes and Structure

Topic: Periodicity

Question Summary:

Elements sodium, silicon, phosphorus, and chlorine are compared in terms of bonding and structure. The question also asks about trends in covalent radius across the period and boiling points of noble gases.

Worked Solution:

(a) Completing the table:

- Sodium: metallic bonding, metallic lattice structure.
- Silicon: covalent bonding, giant covalent network.
- Phosphorus: covalent bonding, molecular (P₄ molecules).
- Chlorine: covalent bonding, diatomic molecular structure (Cl₂).

(b)(i) Trend in covalent radius:

Covalent radius decreases across the period from sodium to chlorine. This is due to increasing nuclear charge pulling electrons closer to the nucleus.

(b)(ii) Why argon has no covalent radius listed:

Argon is a noble gas and does not form covalent bonds under normal conditions, so a covalent radius cannot be measured.

(c) Boiling points of noble gases down group 0:

Boiling points increase down the group because atoms become larger and have more electrons. This increases the strength of London dispersion forces (van der Waals forces). Stronger dispersion forces require more energy to overcome, so boiling points increase.

Revision Tips:

- Remember that metallic structures are lattices of positive ions in a sea of delocalised electrons.
- Giant covalent networks have very high melting and boiling points due to strong bonds throughout the structure.
- Covalent radius trends follow nuclear charge effects.
- London dispersion forces increase with electron number and atom size.